

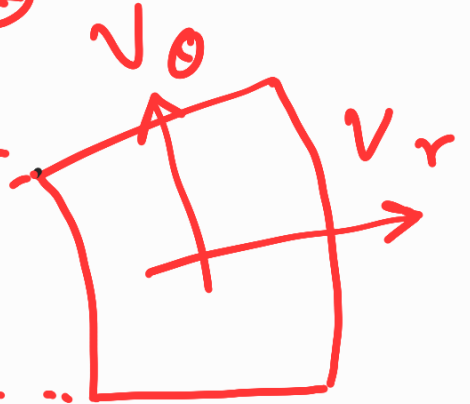
$$\frac{\partial p}{\partial t} + \rho \left[\frac{v_r}{r} + \frac{\partial v_r}{\partial r} \right] + \rho \frac{\partial v_\theta}{\partial \theta} \cdot \frac{1}{r} = 0$$

For incompressible flow (2D)

$$\frac{v_r}{r} + \frac{\partial v_r}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} = 0$$

v_r = radial component
of vel

v_θ = tangential
component of vel.



For 2D vortex flow — no
radial velocity but has
tangential velocity (cylindrical)

ie $v_r = 0$ and $v_\theta \neq 0$

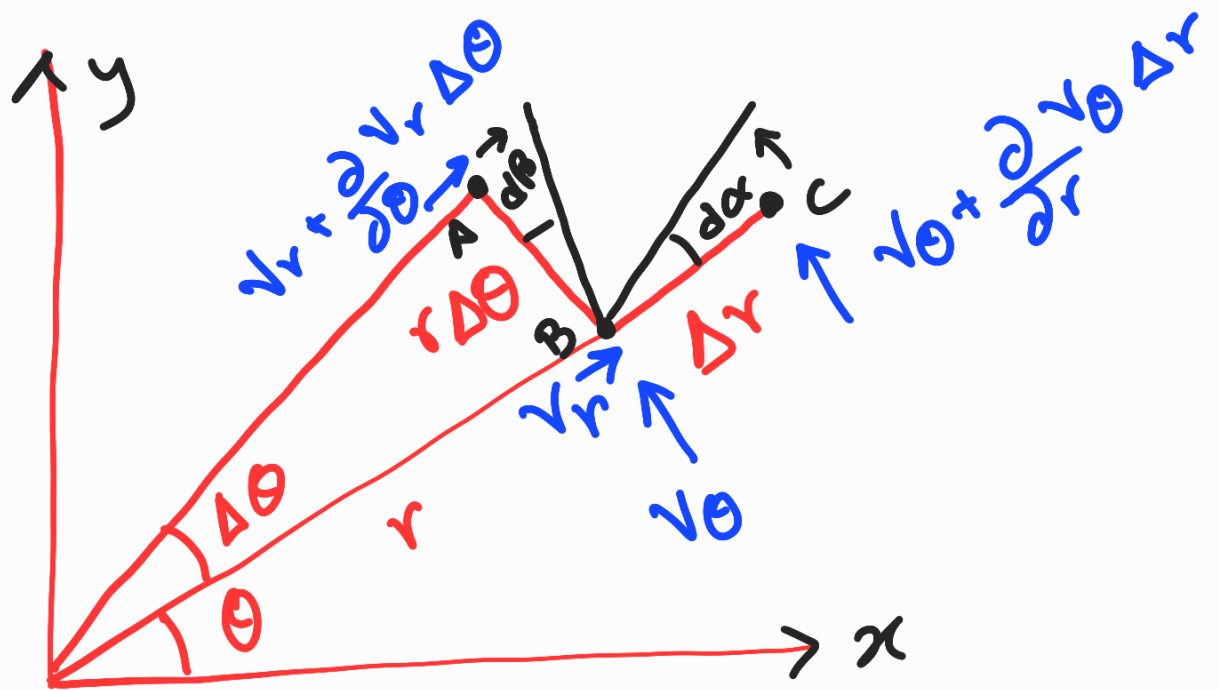
$$\frac{v_r}{r} + \frac{\partial v_r}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} = 0$$

$$\Rightarrow \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} = 0$$

$\Rightarrow v_\theta$ is not function of θ .
 v_θ is function of r only.

$$\Rightarrow v_\theta = v_\theta(r).$$

Rotation in polar coordinate system.



$$\frac{d\alpha}{dt} = \frac{\partial v_\theta}{\partial r} \frac{\Delta r}{\Delta r} = \frac{\partial v_\theta}{\partial r}$$

$$\frac{d\beta}{dt} = \frac{\partial}{\partial \theta} \frac{v_r \Delta \theta}{r \Delta \theta} = \frac{1}{r} \frac{\partial v_r}{\partial \theta}$$

Angular velocity in anticlockwise direction



$= \frac{d\alpha}{dt} + \text{additional angular vel with which the fluid is rotating}$

$$= \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \quad [\because v_\theta = r\omega]$$

Angular vel in clockwise direction



$$= -\frac{1}{r} \frac{\partial v_r}{\partial \theta}$$

$\therefore$ Rotation = avg angular vel

$$\omega_z = \frac{1}{2} \left(\frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} - \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right)$$

For irrotational flow

$$\omega_z = 0 \Rightarrow \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} - \frac{1}{r} \frac{\partial v_r}{\partial \theta} = 0$$

For vortex flow $v_r = 0$ (cylindrical)

$$\Rightarrow \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} = 0$$

$\therefore v_\theta$ is not a function of θ and is only a function of r , ∂ can be replaced by d

$$\Rightarrow \frac{dv_\theta}{dr} + \frac{v_\theta}{r} = 0$$

$$\Rightarrow \frac{1}{r} \frac{d}{dr} (v_\theta \cdot r) = 0$$

$$\Rightarrow v_\theta r = \text{constant}$$

For free vortex flow

$$v_\theta \times r = \text{constant}$$

Hence - free vortex flow is irrotational

For forced vortex flow

$$V_\theta = \omega \times r$$

$$\Rightarrow \frac{V_\theta}{r} = \omega \quad \text{(angular velocity with which the fluid is rotated)}$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial V_\theta}{\partial r} + \frac{V_\theta}{r} - \frac{1}{r} \frac{\partial V_r}{\partial \theta} \right)$$

For vortex flow $V_r = 0$

$$\Rightarrow \omega_z = \frac{1}{2} \left(\frac{\partial V_\theta}{\partial r} + \frac{V_\theta}{r} \right)$$

$$= \frac{1}{2} (\omega + \omega)$$

$$= \omega \neq 0$$

Hence rotational

